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Engineering Applied Mathematics IIB for Three Players with Six Number of Jester Cards on Differential Calculus of Function of Several Variables–Level 4

Dagnachew Jenber

Department of Mathematics, Addis Ababa Science and Technology University, Addis Ababa, Ethiopia.

Abstract

This work presents 24 number of cards from applied mathematics IIB degree level mainly focused on differential calculus of function of several variables. The jester cards are A, B, C, D, E and F.

1 Introduction

At present, the decline in the reading community has reached a worrisome level. For many reasons, people have stopped sitting down and reading. For instance, wasting time on useless social media and in trivial places are among the many minor reasons. In 2025, Jenber proposed a solution to this through a major share company established under the title "Zero or One Trading Share Company." This share company believes that the problem mentioned above can be solved in the following ways. We found the solution included in this writing to be convincing (for more information, see the foundation document of 0 or 1 game [4]). The solution presented by this share company is briefly summarized below:

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(1) Making reading or studying entertaining, profitable, and rewarding. One of the methods for studying or acquiring new knowledge is understanding things by associating them with their equivalents. For example, an English word has many synonyms. Trying to memorize them by associating them is considered one of the best methods. However, to make this exercise unforgettable, doing it in the form of a game while having fun and discussing in groups makes it preferable.There are scientific studies conducted around studying English vocabulary by preparing cards (for example, see [1, 2, 3, 5, 6, 7, 13, 14, 15, 17]).

(2) Understanding things by associating them according to their types stimulates thinking, makes one inquisitive, sparks discussions with friends, and even drives one to search for reference books. And when done this way, it takes a much longer time to forget that thing.

(3) Furthermore, if this association is done enjoyably in the form of a game with friends, and if a bonus/reward is given to the winner to encourage knowledge and learning, the game will become popular.

(4) Developing various types of entertaining games to execute the solutions mentioned in 1-3 above.

In this paper, we have prepared cards for the 0 or 1 game. We have briefly described the number of cards as follows. Let the number of jester cards be 6 and the number of players be $k = 3$. Therefore, the total number of cards, $n = 8 \times 3 + 6 = 30$. Thus, including the jester cards, there are a total of 30 cards. The player cards will be $30 - 6 = 24$ cards; 24 is a multiple of 8 (for more information, see the 0 or 1 game foundation document). The jester cards are A, B, C, D, E and F.

2 Preliminary

Many engineering systems depend on more than one independent variable. For example, temperature may depend on spatial coordinates, electric potential may vary with position, and mechanical stress may depend on several geometric and physical parameters. Differential calculus of functions of several variables provides mathematical tools for describing and analyzing such systems.

The following concepts introduce the main ideas of differential calculus of functions of several variables, with examples related to engineering applications. The discussion follows the multivariable calculus material in [16].

2.1 Functions of Several Variables and Level Curves

Definition. A function of two variables is a rule that assigns a unique real number to each point (x, y) in its domain. It is commonly written as

$$z = f(x, y).$$

The domain is the set of admissible input points, while the range is the set of possible output values.

A level curve of f is a curve on which the function has a constant value c:

$$f(x, y) = c.$$

Example. Suppose the height of a curved surface is described by

$$h(x, y) = x^2 + y^2.$$

The level curve corresponding to $h = 25$ is

$$x^2 + y^2 = 25,$$

which is a circle of radius 5 centered at the origin.

For the function

$$g(x, y) = \sqrt{25 - 4x^2 - y^2},$$

the expression under the square root must be nonnegative. Therefore, its domain is

$$4x^2 + y^2 \leqslant 25.$$

For further discussion, see [16, Section 4.1].

2.2 Limits and Continuity

Definition. The limit of $f(x, y)$ as (x, y) approaches (a, b) is L if $f(x, y)$ approaches L along every path approaching (a, b). We write

$$\lim_{(x,y)\to(a,b)} f(x, y) = L.$$

The function f is continuous at (a, b) if

$$\lim_{(x,y)\to(a,b)} f(x, y) = f(a, b).$$

Example. Consider

$$f(x, y) = \begin{cases} \dfrac{x^2 y}{x^2 + y^2}, & (x, y) \neq (0, 0), \\ 0, & (x, y) = (0, 0). \end{cases}$$

For $(x, y) \neq (0, 0)$,

$$\left| \frac{x^2 y}{x^2 + y^2} \right| \leqslant |y|.$$

As $(x, y) \to (0, 0)$, we have $|y| \to 0$. By the squeeze theorem,

$$\lim_{(x,y)\to(0,0)} \frac{x^2 y}{x^2 + y^2} = 0.$$

Since $f(0, 0) = 0$, the function is continuous at the origin.

See [9, Section 4.2].

2.3 Partial Derivatives and Higher-Order Derivatives

Definition. A partial derivative measures the rate of change of a function with respect to one variable while the other variables are held constant. For $z = f(x, y)$,

$$f_x = \frac{\partial f}{\partial x}, \qquad f_y = \frac{\partial f}{\partial y}.$$

Higher-order partial derivatives are obtained by differentiating again. Examples include

$$f_{xx}, \qquad f_{yy}, \qquad f_{xy}, \qquad f_{yx}.$$

If the mixed partial derivatives are continuous near a point, then

$$f_{xy} = f_{yx}.$$

Example. Suppose the pressure in a chamber is modeled by

$$P(x, y) = x^2 y + 3y^2.$$

Then

$$P_x = 2xy, \qquad P_y = x^2 + 6y.$$

At $(2, 1)$,

$$P_x(2, 1) = 2(2)(1) = 4.$$

This value describes the local rate of change of pressure with respect to x, with y held constant.

See [10, Section 4.3].

2.4 Gradient and Directional Derivatives

Definition. The gradient of a differentiable function $f(x, y)$ is

$$\nabla f(x, y) = \langle f_x(x, y), f_y(x, y) \rangle .$$

For a unit vector $\mathbf{u} = \langle u_1, u_2 \rangle$, the directional derivative of f in the direction of $\mathbf{u}$ is

$$D_{\mathbf{u}} f = \nabla f \cdot \mathbf{u}.$$

The gradient points in the direction of the greatest local increase. The maximum directional derivative is $|\nabla f|$.

Example. Let

$$T(x, y) = x^2 + 3y^2.$$

Then

$$\nabla T = \langle 2x, 6y \rangle.$$

At $(1, 1)$,

$$\nabla T(1, 1) = \langle 2, 6 \rangle.$$

The unit vector in the direction of greatest increase is

$$\frac{\langle 2, 6 \rangle}{\sqrt{2^2 + 6^2}} = \frac{\langle 1, 3 \rangle}{\sqrt{10}}.$$

See [8, Section 4.6].

2.5 Total Differentials and Error Estimation

Definition. For a differentiable function $z = f(x, y)$, the total differential is

$$dz = f_x\,dx + f_y\,dy.$$

For small changes in the independent variables, the differential provides an approximation to the change in the function:

$$\Delta z \approx dz.$$

Example. The diagonal of a rectangular panel is

$$d = \sqrt{x^2 + y^2}.$$

Its total differential is

$$dd = \frac{x}{\sqrt{x^2 + y^2}}\,dx + \frac{y}{\sqrt{x^2 + y^2}}\,dy.$$

This formula estimates how small errors in the measured dimensions affect the calculated diagonal.

See [11, Section 4.4].

2.6 Tangent Planes and Linear Approximation

Definition. The tangent plane to the surface $z = f(x, y)$ at $(a, b, f(a, b))$ is given by

$$\begin{aligned} z = {} & f(a, b) + f_x(a, b)(x - a) \\ & + f_y(a, b)(y - b). \end{aligned}$$

The corresponding linear approximation is

$$\begin{aligned} L(x, y) = {} & f(a, b) + f_x(a, b)(x - a) \\ & + f_y(a, b)(y - b). \end{aligned}$$

Example. Let

$$f(x, y) = x^2 + y^2.$$

At $(1, 2)$, $f(1, 2) = 5$, $f_x(1, 2) = 2$, and $f_y(1, 2) = 4$. Hence the tangent plane is

$$z = 5 + 2(x - 1) + 4(y - 2).$$

See [11, Section 4.4].

2.7 Chain Rule and Implicit Differentiation

Definition. If $z = f(x, y)$, where $x = x(t)$ and $y = y(t)$, the chain rule gives

$$\frac{dz}{dt} = \frac{\partial f}{\partial x}\frac{dx}{dt} + \frac{\partial f}{\partial y}\frac{dy}{dt}.$$

Implicit differentiation is used when variables are related by an equation rather than being explicitly expressed as functions.

Example. Suppose

$$z = x^2 + y^2, \qquad x = 2t, \qquad y = 3t.$$

Then

$$z = (2t)^2 + (3t)^2 = 13t^2,$$

and consequently

$$\frac{dz}{dt} = 26t.$$

See [12, Section 4.5].

2.8 Critical Points and Unconstrained Extrema

Definition. A critical point of $f(x, y)$ is a point where both first partial derivatives are zero or at least one first partial derivative does not exist.

At a critical point, define

$$D = f_{xx} f_{yy} - (f_{xy})^2.$$

The second derivative test gives the following classifications:

- If $D > 0$ and $f_{xx} > 0$, there is a strict local minimum.
- If $D > 0$ and $f_{xx} < 0$, there is a strict local maximum.
- If $D < 0$, the point is a saddle point.
- If $D = 0$, the test is inconclusive.

Example. Consider

$$f(x, y) = x^2 + 2y^2.$$

The origin is a critical point, and

$$D = (2)(4) - 0^2 = 8 > 0, \qquad f_{xx} = 2 > 0.$$

Therefore, $(0, 0)$ is a strict local minimum.

See [16, Section 4.7].

2.9 Constrained Optimization and Lagrange Multipliers

Definition. Constrained optimization finds the maximum or minimum of an objective function subject to one or more restrictions. For the objective $f(x, y)$ subject to $g(x, y) = 0$, the Lagrange multiplier equations are

$$\nabla f = \lambda \nabla g, \qquad g(x, y) = 0.$$

Example. Maximize

$$f(x, y) = xy$$

subject to

$$x + y = 10.$$

Using $g(x, y) = x + y - 10$, the equations are

$$y = \lambda, \qquad x = \lambda, \qquad x + y = 10.$$

Thus $x = y = 5$, and the maximum value is

$$f(5, 5) = 25.$$

See [16, Section 4.8].

3 Jester and Player Cards

3.1 Instructions

Each question has six answer choices, labelled A, B, C, D, E, and F. Exactly one choice is correct for each question, and the other five are incorrect.

During each play, a student selects one answer choice. A correct selection earns a score of 1, while an incorrect selection earns a score of 0.

3.2 Jester Cards

The jester cards contain the texts A, B, C, D, E and F.

3.3 Player Cards

1. An engineering simulation models the height of a curved surface by

   $$h(x, y) = \sqrt{25 - 4x^2 - y^2}.$$

   Which condition describes the domain of this function?

   (a) $4x^2 + y^2 < 25$
   (b) $4x^2 + y^2 \geqslant 25$
   (c) $4x^2 + y^2 = 25$
   (d) $4x^2 + y^2 \leqslant 25$
   (e) $4x^2 - y^2 \leqslant 25$
   (f) $x^2 + y^2 \leqslant 25$

2. A machine component has a modeled stress function

   $$S(x, y) = 2x^2 + y^2.$$

   Which equation describes the level curve corresponding to a stress value of 18?

   (a) $x^2 + 2y^2 = 18$

(b) $2x + y = 18$

(c) $2x^2 + y^2 = 18$

(d) $2x^2 - y^2 = 18$

(e) $2x^2 + y^2 = 9$

(f) $x^2 + y^2 = 18$

3. A normalized sensor response is described near the origin by

$$f(x, y) = \frac{x^2 y}{x^2 + y^2}, \qquad (x, y) \neq (0, 0).$$

What is

$$\lim_{(x,y)\to(0,0)} f(x, y)?$$

(a) 1

(b) -1

(c) $\dfrac{1}{2}$

(d) The limit does not exist.

(e) $+\infty$

(f) 0

4. A sensor's calibration function is defined by

$$f(x, y) = \begin{cases} \dfrac{e^{x+y} - 1}{x + y}, & x + y \neq 0, \\ 1, & x + y = 0. \end{cases}$$

Which statement is correct at $(0, 0)$?

(a) The function is discontinuous because the quotient is undefined.

(b) The function is continuous because the limit equals $f(0, 0) = 1$.

(c) The function has limit 0 but its value is 1.

(d) The function has limit $+\infty$.

(e) The function has no limit because x and y are independent.

(f) The function is continuous only when $x = y$.

5. The response of an electronic device is modeled by

$$R(x, y) = x e^{xy}.$$

What is $\dfrac{\partial R}{\partial x}$ at $(0, 2)$?

(a) 0

(b) 2

(c) e^2

(d) 1

(e) $2e^2$

(f) -1

6. A heat distribution is approximated by

$$T(x, y) = x^2 \sin y + y^3.$$

Find $T_{yy}(1, \pi/2)$.

(a) $3\pi - 1$

(b) $3\pi + 1$

(c) 6π

(d) -1

(e) 3π

(f) $6\pi - 1$

7. A production model uses the function

$$P(x, y) = x^2 y + 3xy^2,$$

where x and y represent two production inputs. Find $P_{xy}(1, 2)$.

(a) 8

(b) 10

(c) 12

(d) 14

(e) 16

(f) 18

8. A curved surface is given by

$$z = f(x, y) = x^2 + xy + y^2.$$

What is the equation of its tangent plane at $(1, 2, 7)$?

(a) $z = 2x + 4y - 5$

(b) $z = 4x + 5y + 7$

(c) $z = 4x + 5y - 7$

(d) $z = 5x + 4y - 7$

(e) $z = x + 2y + 2$

(f) $z = 4x + 5y - 5$

9. A sensor response is modeled by

$$f(x, y) = e^{x+y}.$$

Using linear approximation at $(0, 0)$, estimate $f(0.03, -0.02)$.

(a) 1.05
(b) 0.99
(c) 1.10
(d) 0.95
(e) 1.01
(f) 1.00

10. A mechanical system has an energy function

$$z = x^2 y, \qquad x = t^2, \qquad y = 1 + t.$$

What is $\dfrac{dz}{dt}$ at $t = 1$?

(a) 3
(b) 5
(c) 7
(d) 8
(e) 9
(f) 12

11. A point on a three-dimensional surface satisfies

$$x^2 + 2y^2 + 3z^2 = 6.$$

At $(1, 1, 1)$, find $\dfrac{\partial z}{\partial x}$, treating z as a function of x and y.

(a) $-\dfrac{1}{3}$
(b) $-\dfrac{2}{3}$
(c) -1
(d) $\dfrac{1}{3}$
(e) $\dfrac{2}{3}$
(f) 1

12. The squared distance from a reference point is modeled by

$$f(x, y) = x^2 + y^2.$$

At $(3, 4)$, find the directional derivative in the direction pointing toward the origin.

(a) 50
(b) -25
(c) 0
(d) 25
(e) -40
(f) -10

13. The level curves of a design function are described by

$$f(x, y) = x^2 + xy + y^2.$$

Find the tangent line to the level curve through $(1, 1)$.

(a) $x - y = 0$
(b) $x + y = 2$
(c) $2x + y = 3$
(d) $x + 2y = 3$
(e) $x + y = 3$
(f) $2x - 2y = 1$

14. A scalar field describing a physical quantity is

$$f(x, y) = 2x^2 + y^2.$$

What is the maximum directional derivative at $(1, 2)$?

(a) 4
(b) $4\sqrt{2}$
(c) $2\sqrt{5}$
(d) 8
(e) $4\sqrt{5}$
(f) 10

15. The diagonal of a rectangular panel is

$$d = \sqrt{x^2 + y^2}.$$

At $x = 3\,\mathrm{m}$ and $y = 4\,\mathrm{m}$, the measured dimensions change by $dx = 0.02\,\mathrm{m}$ and $dy = -0.01\,\mathrm{m}$. Estimate the change in the diagonal using the total differential.

(a) $0.020\,\mathrm{m}$

(b) $0.012\,\mathrm{m}$

(c) $-0.004\,\mathrm{m}$

(d) $0.004\,\mathrm{m}$

(e) $0.008\,\mathrm{m}$

(f) $-0.008\,\mathrm{m}$

16. An electrical resistance is calculated from $R = V/I$. A voltage measurement is $V = 12\,\mathrm{V}$ with a maximum absolute error of $0.1\,\mathrm{V}$, and a current measurement is $I = 2\,\mathrm{A}$ with a maximum absolute error of $0.02\,\mathrm{A}$. Using differentials, what is the approximate maximum percentage relative error in R?

(a) 0.833%

(b) 1.000%

(c) 1.833%

(d) 2.000%

(e) 2.833%

(f) 3.000%

17. A moving object's temperature field is modeled by

$$T(x, y) = 100 - 2x^2 - 3y^2, \qquad x = \cos t, \quad y = \sin t.$$

What is $\dfrac{dT}{dt}$ at $t = \pi/4$?

(a) 1

(b) -2

(c) 2

(d) -1

(e) 0

(f) -5

18. A design objective is represented by

$$F(x, y) = x^2 + 4xy + y^2.$$

What is the nature of the critical point at $(0, 0)$?

(a) Strict local minimum

(b) Strict local maximum

(c) Absolute minimum

(d) Absolute maximum

(e) Saddle point

(f) The second derivative test is inconclusive.

19. The operating cost of a system is modeled by

$$C(x, y) = x^2 + y^2 - 4x + 6y.$$

Over all real x and y, what is the minimum value of C?

(a) -13

(b) -9

(c) -4

(d) 0

(e) 5

(f) 13

20. A designer must choose dimensions x and y satisfying $x + y = 6$. The objective is to minimize

$$F(x, y) = x^2 + y^2.$$

What is the minimum value of F?

(a) 6

(b) 12

(c) 18

(d) 24

(e) 30

(f) 36

21. An optimization problem seeks the maximum of

$$f(x, y) = xy$$

subject to $x^2 + y^2 = 8$. What is the maximum value of f?

(a) 2

(b) 4

(c) 8

(d) 16

(e) 0

(f) $8\sqrt{2}$

22. The concentration of a substance is modeled by

$$C(x, y) = e^{-x} \cos(2y).$$

What is $\dfrac{\partial C}{\partial y}$ at $(0, \pi/4)$?

(a) 0

(b) 1

(c) -1

(d) 2

(e) -2

(f) $-2e^{-1}$

23. A displacement magnitude is represented by

$$f(x, y) = \sqrt{x^2 + y^2}.$$

Which statement correctly describes the function at the origin?

(a) It is neither continuous nor differentiable.

(b) It is differentiable but not continuous.

(c) It is continuous and differentiable.

(d) It is continuous but not differentiable.

(e) It has no limit at the origin.

(f) It is discontinuous only along the x-axis.

24. A rectangular box has positive dimensions x, y, z, with

$$x + y + z = 12.$$

Its volume is $V = xyz$. What is the maximum possible volume?

(a) 16

(b) 27

(c) 36

(d) 48

(e) 60

(f) 64

Scoring rule:

$$\text{Score} = \begin{cases} 1, & \text{if the selected choice is correct,} \\ 0, & \text{if the selected choice is incorrect.} \end{cases}$$

3.4 Answer Key

Question	Answer	Question	Answer	Question	Answer
1	D	9	E	17	D
2	C	10	E	18	E
3	F	11	A	19	A
4	B	12	F	20	C
5	D	13	B	21	B
6	A	14	B	22	E
7	D	15	D	23	D
8	C	16	C	24	F

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